

Rumour Detection on Twitter Cascades: A Graph Neural Network Approach with Topological and XAI Analysis

Nuno Batista and Miguel Cabral Pinto

University of Coimbra, Coimbra, Portugal

Abstract. Automated rumour detection on social media requires classifiers that generalise across unseen news events, yet most prior work evaluates within a single train/test split. We study this challenge on the PHEME dataset of 6,178 Twitter cascade graphs spanning seven real-world events, using a strict two-phase leave-one-event-out protocol that keeps two events entirely withheld during hyperparameter search. We benchmark Random Forest, MLP, GCN, and GAT under three feature ablations (full, text-only, structural-only), and complement the classification study with a network science characterisation of cascade topology and GNNExplainer and GAT attention interpretability analyses. Structural-only models fall below the random lower bound on the held-out test set (macro F1 < 0.375), confirming that pure topology does not generalise across events, despite consistent structural differences between classes (rank-biserial $r < 0.15$ throughout). Text-based models all exceed the lower bound; GAT with text features achieves the best combined macro F1 of 0.610, significantly outperforming flat mean-pooled MLP (McNemar $p < 0.001$). Adding structural features to text-based models consistently degrades held-out performance, indicating topology is event-specific rather than event-general. GNNExplainer reveals that authority signals (follower count, PageRank) are more important for rumour predictions than non-rumour ones, while GAT attention concentrates on fewer nodes for rumours, consistent with an anomalous-node detection pattern.

1 Introduction

Misinformation on social media poses a challenge that scales with platform size. Vosoughi et al. [11] found that false news propagates roughly six times faster than true news on Twitter, reaching far larger audiences before corrections can circulate. The consequences, from electoral interference to vaccine hesitancy, have sustained a substantial body of automated detection research.

A Twitter reply thread is a natural directed graph: the source tweet is the root and each reply connects to its parent, forming a cascade tree that encodes both content and propagation dynamics. Most early detection systems treat rumour classification as a flat text problem, aggregating per-tweet features into a single vector and discarding the relational structure. Graph Neural Networks

(GNNs) [5, 10] make it possible to learn directly over these cascade graphs. Prior work, notably BiGCN [2] (which propagates information bidirectionally through the cascade tree) and geometric deep learning on propagation graphs [8], has shown that exploiting graph structure improves over flat baselines. However, these studies typically evaluate within a single train/test split, leaving open whether GNNs learn event-transferable cascade signatures or whether apparent gains reflect overfitting to training-event topology distributions.

This paper addresses three questions using the PHEME dataset [13], a collection of 6,425 Twitter cascade graphs across nine real-world news events labelled as Rumour or Non-Rumour. First, does graph topology alone generalise across unseen events? Second, does message passing add value beyond mean-pooled text embeddings when node features are held constant? Third, which features drive GNN predictions, and do these patterns differ between classes? We adopt a strict leave-one-event-out (LOEO) evaluation protocol, keeping two events entirely held out from hyperparameter search.

Our contributions are threefold. We characterise cascade topology across six analyses spanning cascade size and depth, structural virality [4], network robustness, echo chambers, and novel branching metrics, finding consistent but small structural differences: rumour cascades are shallower and more broadcast-like, with lower Strahler branching complexity, yet these effects are insufficient to drive cross-event classification (structural-only models fall below the random lower bound). We then benchmark MLP, GCN, and GAT under three feature ablations (full, text-only, and structural-only) with bootstrap confidence intervals and McNemar’s significance tests; GAT with text features achieves a combined macro F1 of 0.610, significantly outperforming flat mean-pooled MLP ($p < 0.001$), confirming that message passing provides genuine signal beyond text alone. Finally, we apply GNNExplainer [12] and GAT attention analysis: GNNExplainer reveals the GCN assigns significantly higher feature importance to authority signals (follower count, PageRank) for rumour graphs, while GAT attention is highest at depth-2 in both classes and shows lower per-graph entropy for rumours, indicating more concentrated attention.

2 Dataset and Problem Formulation

2.1 The PHEME Dataset

We formulate rumour detection as binary graph-level classification. Given a cascade $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ (nodes = tweets, edges = replies, $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times d}$ the node feature matrix), the goal is to predict $y \in \{0, 1\}$ (Non-Rumour / Rumour) for the graph as a whole.

We use the PHEME dataset [13], comprising nine real-world Twitter news events with thread-level Rumour/Non-Rumour annotations. Two events are excluded due to near-degenerate class distributions: *ebola-essien* (100% Rumour, 14 cascades) and *prince-toronto* (98.3% Rumour, 233 cascades). A further event, *ferguson*, is excluded from classifier evaluation owing to distributional shift that

Table 1. PHEME event statistics after excluding *ebola-essien* and *prince-toronto*. Events marked (†) form the held-out test set; (‡) is excluded from classifier evaluation (distribution shift) and used only in the network-science analysis.

Event	Graphs	Rumour	Non-Rumour	R%
charliehebd [†]	2,079	458	1,621	22.0
ferguson [‡]	1,143	284	859	24.8
germanwings	469	238	231	50.7
gurlitt	138	61	77	44.2
ottawashooting	890	470	420	52.8
putinmissing	238	126	112	52.9
sydneyseige [†]	1,221	522	699	42.8
Total	6,178	2,159	4,019	34.9

substantially inflates cross-event variance. The remaining seven events yield 6,178 graphs in total; event-level statistics are shown in Table 1. The positive rate varies from 22.0% (charliehebd) to 52.9% (putinmissing), reflecting genuine differences in how each news event unfolded on Twitter. We apply a minimum-size filter of $|\mathcal{V}| \geq 3$, removing cascades that consist only of a source tweet and at most one reply.

Class imbalance is handled at training time by per-event majority-class undersampling, ensuring that no single event’s distribution dominates the training set.

As an additional contribution, we convert the PHEME annotations and graph structures into a ready-to-use PyTorch Geometric dataset and release it publicly on Zenodo [1] and Hugging Face¹ for convenient programmatic access, removing the need to parse raw JSON files to reproduce our experiments.

2.2 Graph Construction and Node Features

Graph construction. We parse each PHEME thread directory into a directed graph: the source tweet becomes the root node (index 0), and each reaction tweet adds a directed edge from its parent (identified via `in_reply_to_status_id`) to itself. Reactions whose parent is absent from the cascade are connected directly to the root, a common artefact of Twitter data collection. The result is a rooted directed tree where edges point from parent to child.

Node features. Each node carries a $d = 390$ -dimensional feature vector: 384-dim text embeddings from `all-MiniLM-L6-v2` [9], 2 user features ($\log(1+\text{followers})$ and verified status), and 4 structural features (PageRank, degree, in-degree, out-degree centrality). Two positional features, `is_root` and `BFS_depth`, are appended at load time, bringing the full dimension to $d = 392$. We evaluate three feature configurations: *full* ($d = 392$), *text-only* ($d = 386$), and *structural-only* ($d = 6$).

¹ <https://huggingface.co/datasets/NunoBatista/PHEME-Misinformation-Graphs>

2.3 Evaluation Protocol

We use a two-phase LOEO design to prevent hyperparameter leakage from the test events (charliehebdo and sydneyseige, marked † in Table 1). In the *search phase* these two events are withheld entirely; 50 guided trials per model-configuration pair are run over the four training events with macro F1 as the criterion. In the *evaluation phase*, models are retrained on all four training events and evaluated on the 3,106-graph combined test set. We report macro F1, precision, recall, and accuracy, together with bootstrap 95% confidence intervals and McNemar’s test for pairwise comparisons.

3 Methodology

3.1 Baseline Models

Random Forest (RF). Each cascade is reduced to a single vector by mean-pooling its node feature matrix: $\mathbf{g} = \frac{1}{|\mathcal{V}|} \sum_v \mathbf{x}_v$. A Random Forest classifier (100 trees, balanced class weights) is trained on these vectors. Any GNN that outperforms the RF gains something specifically from message passing rather than from richer pooling.

Multi-Layer Perceptron (MLP). The MLP also operates on mean-pooled vectors, using fully connected layers with ReLU activations and dropout, followed by a single output logit. It shares the same input as the RF but is differentiable and depth-tunable, isolating the contribution of a learned non-linear classifier before any graph structure is introduced.

3.2 Graph Neural Network Models

Both GNN architectures share three design choices that go beyond their respective base convolutions: residual skip connections, layer normalisation, and a root readout that concatenates the mean-pooled graph embedding with the root node’s final representation.

Graph Convolutional Network (GCN). We use the spectral graph convolution of Kipf and Welling [5]. At each layer ℓ , the representation of node v is updated as:

$$\tilde{\mathbf{h}}_v^{(\ell)} = \sum_{u \in \mathcal{N}(v) \cup \{v\}} \frac{1}{\sqrt{\tilde{d}_v} \sqrt{\tilde{d}_u}} \mathbf{h}_u^{(\ell-1)} \mathbf{W}^{(\ell)} \quad (1)$$

where $\tilde{d}_v = |\mathcal{N}(v)| + 1$ is the degree after adding self-loops. The pre-activations are passed through a residual block, $\mathbf{h}_v^{(\ell)} = \text{LayerNorm}(\text{ReLU}(\tilde{\mathbf{h}}_v^{(\ell)} + \mathbf{W}_{\text{skip}}^{(\ell)} \mathbf{h}_v^{(\ell-1)}))$, where $\mathbf{W}_{\text{skip}}^{(\ell)}$ is a bias-free linear projection. LayerNorm stabilises training on the shallow cascades common in PHEME.

Graph Attention Network (GAT). We use the GATv2 attention mechanism of Brody et al. [3], which addresses a rank-collapse limitation of the original GAT [10] by computing a fully dynamic attention score:

$$e_{ij} = \mathbf{a}^\top \text{LeakyReLU}\left(\mathbf{W} [\mathbf{h}_i^{(\ell-1)} \parallel \mathbf{h}_j^{(\ell-1)}]\right) \quad (2)$$

Scores are normalised via softmax over the neighbourhood, $\alpha_{ij} = \exp(e_{ij}) / \sum_{k \in \mathcal{N}(i)} \exp(e_{ik})$. With K parallel heads, representations are concatenated and passed through the same residual block as the GCN, with $\mathbf{W}_{\text{skip}}^{(\ell)}$ projecting to $K \times d_\ell$. The number of heads K is tuned during hyperparameter search.

Shared readout. After L message-passing layers, both models produce a graph-level representation by concatenating the mean-pooled node embeddings with the root node’s final embedding:

$$\hat{\mathbf{g}} = \left[\frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \mathbf{h}_v^{(L)} \parallel \mathbf{h}_0^{(L)} \right] \quad (3)$$

The root node (index 0, the source tweet) carries information about the cascade’s origin that mean pooling would dilute. A single linear layer maps $\hat{\mathbf{g}}$ to the output logit: $\hat{y} = \sigma(\mathbf{w}_c^\top \hat{\mathbf{g}})$.

Edge direction. The raw cascade graphs are directed (parent $\rightarrow$ child). We experiment with three configurations: *original* (edges point from source to reply, information flows bottom-up through the classifier), *inverted* (reversed), and *bidirectional* (both directions added via `to_undirected`, effectively doubling the edge count). Edge direction is treated as a discrete hyperparameter during search.

3.3 Training Procedure

Optimiser and schedule. All neural models are trained with the Adam optimiser and a cosine annealing learning rate schedule with minimum learning rate $\eta_{\min} = \eta_0/10$, where η_0 is the initial rate. Gradients are clipped to ℓ_2 norm 1.0 at every step to prevent instability on very small graphs.

Loss function. We use binary focal loss [6]:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N (1 - p_{t,i})^\gamma \log p_{t,i} \quad (4)$$

where $p_{t,i} = \sigma(\hat{y}_i)$ for positive examples and $p_{t,i} = 1 - \sigma(\hat{y}_i)$ for negative examples. The modulating factor $(1 - p_{t,i})^\gamma$ down-weights well-classified examples and concentrates learning on hard or misclassified ones, which is particularly beneficial given the class imbalance present in several events. The exponent γ is tuned during hyperparameter search.

Class balancing. Before each training run, the training set is balanced by per-event majority-class undersampling: within each event, the larger class is randomly subsampled to match the size of the smaller one. This prevents events with heavily skewed class ratios from dominating the training signal.

Hyperparameter search. Hyperparameters are selected with a sequential model-free search: 50 guided trials are run per model-configuration combination (12 combinations in total: 4 models $\times$ 3 feature configurations), where each trial evaluates the full LOEO procedure over the four training events and reports the global macro F1. The search space covers hidden layer dimensions, dropout rate, learning rate, weight decay, focal loss exponent γ , number of attention heads (GAT only), and edge direction (GCN and GAT). The best configuration per combination is used for all final held-out evaluations.

4 Related Work

GNN-based rumour and fake news detection. Graph-based approaches to rumour detection represent information cascades as graphs and apply message-passing to exploit propagation structure. Bian et al. [2] introduced BiGCN, which propagates representations bidirectionally along the cascade tree (top-down and bottom-up), consistently outperforming sequential baselines on Twitter15/16. Lu and Li [7] proposed GCAN, which jointly models user interaction graphs and source-content co-attention, also providing post-hoc explanations via attention weights. Monti et al. [8] applied spectral graph convolutions to the propagation graphs of FakeNewsNet, demonstrating that geometric deep learning transfers across datasets. Relative to these works, our study makes three methodological departures: we use a strict two-phase LOEO protocol that keeps two events entirely hidden during hyperparameter search; we perform a controlled feature ablation that isolates the contribution of message passing from that of the text embedding; and we complement classification with a network-science characterisation of cascade topology.

Explainability and cascade topology. GNNExplainer [12] provides post-hoc explanations by optimising a soft mask over the node features and subgraph of each prediction; we use it to compare the feature-group importance the model assigns to rumour versus non-rumour graphs. For cascade topology, Goel et al. [4] defined structural virality as the average pairwise distance in a diffusion tree, providing a principled measure of broadcast versus chain-like spread that we adopt as one of our six topological analyses.

5 Network Science Analysis

We characterise the structural properties of the full 6,425-graph dataset (all events, no size filter) across six analyses. All comparisons use two-sided Mann-Whitney U tests. The rank-biserial effect size $r \in [0, 1]$ measures practical magnitude as the non-parametric correlation between class membership and the metric

(0 = no separation, 1 = perfect separation); thresholds: small $r \geq 0.1$, medium $r \geq 0.3$. Cascade size and depth additionally report Spearman ρ , the rank-order correlation between metric value and the binary Rumour/Non-Rumour label. The central finding is that rumour cascades are consistently shallower and more broadcast-oriented than non-rumour cascades, but all effects are small ($r < 0.15$), and the structural signal does not generalise across events (confirmed in Section 6).

Cascade size and depth. Rumour cascades are significantly smaller (mean 13.89 nodes vs. 18.00; Spearman $\rho = -0.119$, $p < 0.001$) and shallower (mean depth 1.257 vs. 1.528; $\rho = -0.102$, $p < 0.001$). Both effects are statistically robust at $n = 6,425$ but small in magnitude.

Tipping point. Stratifying by cascade size reveals a tipping-point pattern: rumour prevalence declines monotonically from 44.9% among cascades with 1–5 nodes to 11.1% in those with more than 200 nodes, consistent with factual content continuing to accumulate replies while rumour threads stall early.

Structural virality. Following Goel et al. [4], structural virality is defined as the average pairwise path length (APL) in the cascade tree. Rumour cascades have significantly lower APL (mean 2.463 vs. 2.808, $p < 0.001$, $r = 0.114$), confirming a structurally more star-like topology (replies cluster around the root) while non-rumour cascades exhibit longer information chains consistent with viral diffusion. The effect is small.

Network robustness and echo chambers. Robustness is measured by tracking the largest connected component fraction after removing the top 1%–30% of degree-ranked hubs. Both classes show near-identical curves, differing by at most 0.014 at any removal rate, with the curves crossing at 10% removal. The branching factor (mean out-degree of internal nodes) is also indistinguishable between classes ($p = 0.650$, $r = 0.007$).

Novel topological metrics. Table 2 reports six cascade complexity metrics. Four discriminate significantly. Root Dominance (fraction of all edges incident on the root node) is higher for rumours ($r = 0.074$), reinforcing the broadcast/star pattern. Strahler Number, a recursive measure of tree branching complexity analogous to stream order in hydrology, is lower for rumours ($r = 0.104$), indicating structurally simpler cascade trees. Normalised Depth (max depth divided by $\log_2 n$) is lower for rumours ($r = 0.070$), showing that rumour cascades are shallower than expected for their size. The Gini coefficient of the out-degree distribution (0 = all nodes have equal out-degree; 1 = one node holds all edges) is lower for rumours ($r = 0.065$), meaning non-rumour cascades contain more high-degree hub nodes. Leaf Ratio (fraction of nodes with out-degree zero, i.e., no replies) is marginally significant ($p = 0.010$, $r = 0.039$). Epidemic R_0 , the mean out-degree of nodes that themselves received at least one reply (analogous to the basic reproduction number in epidemiology), does not reach significance ($p = 0.081$).

Table 2. Novel topological metrics comparing Rumour (R) and Non-Rumour (NR) cascades ($n = 6,425$, Mann-Whitney U). r = rank-biserial effect size. Significance: *** $p < 0.001$, ** $p < 0.01$, ns $p \geq 0.05$.

Metric	R mean	NR mean	p -value	r
Root Dominance	0.765	0.727	< 0.001	0.074 ***
Gini out-degree	0.683	0.718	< 0.001	0.065 ***
Strahler Number	1.893	2.024	< 0.001	0.104 ***
Leaf Ratio	0.736	0.722	0.010	0.039 **
Norm. Depth	0.726	0.822	< 0.001	0.070 ***
Epidemic R_0	4.214	4.469	0.081	0.028 ns

Table 3. Final held-out evaluation. Models are trained on four events and tested on charliehebdo and sydneyseige (never seen during hyperparameter search). Combined macro F1 over 3,106 graphs; Δ is the margin above the random lower bound (0.375). Best result per column in **bold**.

Model	Config	Charlie F1	Sydney F1	Comb. F1	Acc	Δ
GAT	text	0.552	0.653	0.610	0.770	+0.235
GAT	full	0.557	0.576	0.565	0.709	+0.190
GCN	text	0.504	0.604	0.551	0.686	+0.176
GCN	full	0.540	0.557	0.548	0.697	+0.173
MLP	text	0.473	0.577	0.520	0.643	+0.145
MLP	full	0.460	0.600	0.521	0.614	+0.146
MLP	struct	0.365	0.000	0.303	0.353	−0.072
GCN	struct	0.289	0.282	0.287	0.414	−0.088
GAT	struct	0.152	0.247	0.200	0.666	−0.175

6 Classification Results

We evaluate four classifiers under three feature configurations on the fixed held-out test set (charliehebdo, 1,952 graphs; sydneyseige, 1,154 graphs; 3,106 combined). The random-classifier lower bound (predict-all-positive at $p = 0.5$) gives a combined macro F1 of 0.375. All neural models are trained with the best hyperparameters found during the search phase, which excluded the two test events entirely.

Table 3 reports per-event and combined macro F1 for the six relevant model-configuration pairs.

Structural-only models. All three structural-only classifiers fall below the random lower bound (combined F1 of 0.200–0.303 vs. 0.375). This confirms the finding from Section 5: the topological differences identified between classes are real but too small to support cross-event generalisation.

Text and full-feature models. All six text and full-feature models exceed the lower bound by comfortable margins (+0.145 to +0.235). GAT with text features

achieves the best combined F1 of 0.610. Adding structural and positional features to the text-based models consistently hurts generalisation: both GAT (0.610 $\rightarrow$ 0.565) and GCN (0.551 $\rightarrow$ 0.548) perform worse in the full configuration. The structural features overfit the training event distributions and do not transfer to the two test events.

Statistical tests. Bootstrap 95% CIs for the top four models (GAT-text [0.583, 0.636], GAT-full [0.540, 0.590], GCN-text [0.526, 0.576], GCN-full [0.521, 0.573]) all exclude the random lower bound, confirming significance above chance. All pairwise comparisons on the combined 3,106-prediction test set are significant at $p < 0.001$. Message passing beats flat MLP in both feature settings, confirming that cascade graph structure provides genuine signal beyond mean-pooled embeddings. Attention (GAT) beats GCN in both settings, justifying the added mechanism complexity. Adding structural features significantly hurts both GCN and GAT, indicating that topology encodes event-specific rather than event-general patterns. The GNN-MLP F1 margin is modest (+0.031 for text, +0.027 for full), suggesting NLP embeddings carry the bulk of the discriminative signal.

7 XAI Analysis

To understand what drives the GNN predictions, we apply two complementary explainability methods on a separate training run using the charliehebd event as the held-out test fold (1,952 graphs: 436 Rumour, 1,516 Non-Rumour). The models trained for XAI are not the HP-searched models from Section 6; they use default hyperparameters and are trained on the remaining events. GNNExplainer [12] is applied to the GCN, and attention analysis is applied to the GAT.

GNNExplainer: feature group importance. GNNExplainer [12] optimises a soft per-node feature mask to identify which features drove each prediction. We apply it to 40 balanced test graphs, average mask weights within semantic feature groups, and compare per-graph group importance between classes via Mann-Whitney U; higher importance means the model relied more on that group. The GCN relies significantly more on every feature group when processing Rumour graphs than Non-Rumour graphs. The largest differences are in authority signals: follower count ($r = 0.508$, $p < 0.001$) and PageRank ($r = 0.513$, $p < 0.001$). The NLP embedding group is also more important for Rumour graphs ($r = 0.464$, $p < 0.001$), as are degree and in-degree centrality ($r \approx 0.49$, $p < 0.001$). Out-degree is the only non-significant group ($p = 0.889$), suggesting the number of replies a user generates is not discriminative. Rumour cascades are thus flagged by a combination of anomalous authority signals and unusual text, rather than any single feature.

GAT attention by cascade depth. The second GAT layer assigns each edge an attention weight α_{uv} (Eq. 2); we aggregate these by the BFS depth of the receiving node (depth 0 = root, depth 1 = direct replies, etc.) and compare across

classes. Depth-2 nodes receive disproportionately high attention in both classes (mean 0.719–0.776 vs. approximately 0.50 at all other depths). Non-Rumour cascades receive significantly more depth-2 attention than Rumour cascades (0.776 vs. 0.719, $p < 0.001$, $r = 0.233$), suggesting non-rumour threads generate richer second-tier engagement. The pattern reverses at depth 3 (Rumour $>$ Non-Rumour, $p = 0.001$, $r = 0.106$); beyond depth 3 differences are not significant.

Attention entropy. Per-graph attention entropy measures how diffuse or concentrated attention is (higher = spread evenly across nodes; lower = focused on a few nodes). Rumour cascades have significantly lower entropy (mean 2.394 vs. 2.615 for Non-Rumour, $p < 0.001$), indicating that the GAT concentrates on fewer nodes when classifying rumours. This aligns with the GNNExplainer finding: rumour cascades appear to contain a small number of structurally or semantically anomalous nodes (high-authority users, unusual depth-2 reply patterns) that dominate the model’s decision, whereas non-rumour engagement is more evenly distributed.

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